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Affect recognition on infants

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ABSTRACT

Before infants learn to speak, they rely heavily on facial expressions to communicate. Automated recognition of infant expressions can be a challenge due to subtle and co-occurring facial movements, limited data and requirement of manual coding that is labor-intensive. This study evaluated two automated approaches for classifying infant affect from facial images: one based on Action Unit (AU) detection using the PyAFAR INFANT model, and another using a fine-tuned Convolutional Neural Network (CNN) pre-trained on emotion recognition. Using AUs as features, a Support Vector Machine (SVM) classifier achieved 0.86 accuracy, but showed lower performance in detecting negative affect (0.76), with misclassifications driven by co-activation of smile and cry-related AUs, and under-detections of brow-related AUs. A CNN model improved overall performance, achieving 0.96 accuracy. A comparison between SVM models using common AUs detected by the infant-trained and adult-trained PyAFAR models shows that the adult model performed significantly worse, emphasizing the importance of relying on infant-trained models. These results demonstrate that AU-based and fine-tuned CNN approaches can effectively be used for infant affect recognition, with CNN proving better performance, particularly in handling ambiguous and subtle expressions.

CONTENTS

1 INTRODUCTION

Being able to understand the affect of an infant is important as infants primarily communicate nonverbally before developing any kind of speech mechanism. Affect is a psychological term used to describe the outward expression of emotion and feelings [Tao and Tan, 2005]. Infants show their emotions and socially interact through facial expressions, body movements, and simple vocalization. Face and facial expressions are undoubtedly one of the most important nonverbal channels used. Emotional recognition begins to develop around six months of age, which is vital for social referencing. This means that infants "read" their parents' expressions to understand what is happening around them [Walker-Andrews, 1998]. Parents and caregivers need to be able to process infants' facial expressions to be able to respond appropriately as well as properly care for them. Understanding infant affect is important for both developmental research and clinical applications, as it can inform responsive caregiving and early identification of socio-emotional difficulties.

To systematically measure facial expressions and provide a framework for doing so for others, Paul Ekman and Wallace V. Friesen published Facial Action Coding System (FACS) [Ekman and Friesen, 1978], it has had a significant update [Ekman et al., 2002], and had an infant-specific extension BabyFACS [Oster, 2006]. Action units (AUs) are the movement of a facial muscle or muscle groups that configure the expression of an affect or emotion, based on [Ekman and Friesen, 1978]. FACS defines discrete AUs that correspond to specific muscle movements, and BabyFACS adapts these AUs to the infant face, covering almost all observable facial movements.

The prototypical expression of positive affect in infancy is through smiles, which are indexed by raising of the lip corners via the zygomaticus major (AU12) [Messinger et al., 2001]. Infant negative affect is predominantly displayed through frowns and cry-faces, indicated by stretching of the lips via the risorius (AU20) [Messinger et al., 2011, Yale et al., 2003], often combined with brow lowering (AU3, AU4) and lip movements like pouting (AU17) [Messinger et al., 2012]. Eye constriction (AU6) and mouth opening (AU25, AU26, AU27) are also associated with the intensity of both positive and negative expressions [Kohut et al., 2012, Izard, 1987]. As shown in Table 1, these configurations have also been found in other studies [Zaharieva et al., 2023].

Table 1: Expected main and interaction effects of action unit configurations predicting manual codings of positive versus negative facial expressions (adapted from [Zaharieva et al., 2023])

Facial expression valence	**Automatically detected AUs**
Positive facial expressions	
Smiles with eye constriction	Main effect of AU12
Smiles with mouth opening	AU12 * AU(25 + 26 + 27)
Negative facial expressions	
Pouting with brow lowering	Main effect of AU20
Lip stretching with brow lowering	AU20 * AU(3 + 4)
Lip stretching with eye constriction	AU20 * AU(6 + 7)
Lip stretching with mouth opening	AU20 * AU(25 + 26 + 27)

Note. Configurations involving eye constriction [AU6] (referred to as "cheek raising" in the FACS manual [Ekman et al., 2002]) are Duchenne expressions (i.e., Duchenne smiles and Duchenne cry-faces).

Infant faces also differ from adult faces in terms of proportion, e.g. larger eyes and smaller jaw-to-face ratio, they have smoother skin, as well as fewer wrinkles and furrows [Oster, 2006]. Manual coding using BabyFACS can be extremely time-consuming and requires skilled and/or trained professionals to perform the work. It can be labor-intensive [Cohn and Ekman, 2005], making it impractical as the datasets grow in size, or if real-time assessment is necessary. This limitation is growing interest in automated solutions.

Unlike in infants, automated detection of AUs in adult faces has been widely studied. To date, there have been only a handful of studies investigating infant affect using facial expressions in very large samples. Automated AU detection promises to speed up the data processing and improve objectivity. In the infant domain, one recent approach is PyAFAR's Infant module (Automated facial recognition in infants) [Ertugrul et al., 2024], later referred to as PyAFAR INFANT, which uses Histogram of Gradients (HoG) features with Light Gradient Boosting Machines (lightGBM) classifiers for each AU. PyAFAR INFANT also allows for the detection of AUs over time through video sequences.

In this study, we try to investigate whether the AUs estimated by PyAFAR INFANT can be used as features to classify infant affect as positive, neutral, or negative. We also compare this AU-based approach to a deep learning alternative, using transfer learning on a

CNN pre-trained on emotion recognition, which uses images as input. Our main research questions are as follows:

- **How well can the Action Units estimated by PyAFAR INFANT be used for classifying infant affect?**
  - What are the limitations of PyAFAR INFANT in Action Unit detection to predict affect?
  - How does its performance compare to AUs estimated by PyAFAR ADULT?
- **Can a deep learning approach operating directly on images outperform Action Unit based features models in classifying infant affect?**

1.1 RELATED WORK

Early efforts to automate infant facial expression analysis showed that machine-based measurements could effectively replace manual coding. For instance, [Messinger et al., 2009] used an automated facial analysis system to track facial movements like smile strength and eye constriction in infant-mother interactions. By modeling both facial shape and appearance, the system captured expressions continuously over time. These automated measurements closely matched expert FACS/BabyFACS coding and observer ratings of positive emotion, providing early evidence that infant facial affect can be objectively and reliably quantified.

Several open-source toolkits with various pipelienes exist for facial action unit and expression recognition, including OpenFace and Py-Feat [Baltrusaitis et al., 2018, Cheong et al., 2021]. These systems offer AU detection for adult faces and, in Py-Feat's case, basic emotion classification. However, they are trained exclusively on adult datasets and were not reproducible in our environment due to outdated dependencies. As a result, they are poorly suited for infant AU detection, highlighting the motivation for this study and the development of a robust, publicly available alternative in the form of PyAFAR.

Commercial software has also emerged for infant emotion recognition. One with promising results and several applications is Baby FaceReader [Zaharieva et al., 2023], which applies deep neural networks to detect the presence and intensity of AUs. In [Zaharieva et al., 2023] study with infants at 4 and 8 months old, Baby FaceReader's global valence score - a continuous measure estimating the infant's overall emotional tone on a scale from negative to positive - achieved an AUC of 0.81 when distinguishing positive from negative/neutral facial expressions, and an AUC of 0.58 for the more challenging task of distinguishing negative from neutral expressions. A simpler approach using the automatically detected smiling AU (AU12) alone showed the same performance (AUC = 0.86). Additionally, brow lowering (AU3 + AU4) showed promise for identifying negative expressions from neutral ones (AUC = 0.79). These findings suggest that while some AUs are informative for recognizing affects, the performance of the tool remains limited to distinguish negative affects. [Zaharieva et al., 2023] results have highlighted which AUs are most informative: AU12 (lip corner puller) by itself was a strong indicator of positive affect, whereas the combination of AU4 (brow lowerer) and AU20 (lip stretcher) proved important for detecting negative affect. Furthermore, the rate of 'classification failed' for Baby FaceReader reached 26% for 8-month-olds, raising concerns about robustness in more dynamic recordings. Baby FaceReader offers a user-friendly interface, but its closed-source nature limits transparency and scientific reproducibility, and comparability.

This difficulty in recognizing negative affect is not unique to Baby FaceReader. Similar limitations have been reported in other studies. For instance, Infant AFAR models also show reduced accuracy when detecting negative facial actions such as cry-faces, likely due to the brief and subtle nature of these expressions [Onal Ertugrul et al., 2023]. Likewise, [Hammal et al., 2015b] achieved only moderate classification rates (65%) for positive vs. negative infant affect using dynamic head and facial movements, highlighting the broader challenge of capturing negative emotional states in infants. [Messinger et al., 2012] research further highlights these challenges. In a study examining the combination of facial actions in infants, it was found that certain facial actions, like eye constriction and mouth opening, are associated with both positive and negative expressions, making it difficult to distinguish between the two based solely on these cues. Additionally, automated measurements of facial expressions in infant-mother interactions showed high associations with positive emotion ratings, but were less effective in capturing negative affect, suggesting limitations in both automated and manual coding approaches when it comes to negative emotions [Messinger et al., 2009].

To address the limitations of the adult-trained AU detectors on infant faces, [Hinduja et al., 2023] developed Infant AFAR, a model fine-tuned on infant-specific data, which outperformed previous approaches. By using transfer learning and freezing lower-level layers pre-trained on ImageNet [Deng et al., 2009] that contains 1.2 million images, the CNN learned features suited to infant faces by training the adapted network on MIAMI [Hammal et al., 2015a] and CLOCK [Bian et al., 2023] datasets. It detected a limited but relevant set of AUs central to infant affect (AU1, AU2, AU3, AU4, AU6, AU9, AU12, AU20, AU28). Despite covering fewer AUs than some commercial tools, it has demonstrated high accuracy and generalizability. PyAFAR have adapted their architecture along the years, with the latest infant model using Histogram of Gradients (HoG) features to ensure better privacy and trains a LightGBM model for classification [Ertugrul et al., 2024], that is also the latest version of PyAFAR INFANT that we focus on.

Another important thread of research has explored relationships between AUs to improve the recognition of complex facial expressions. Facial actions often occur in meaningful combinations, especially in infants. [Zaker et al., 2021] addressed this issue by recognizing combinations of AUs and by modelling the dependencies between AUs. They investigated the association among AUs central to positive and negative emotion and proposed a methodology for jointly detecting correlated AUs, particularly the joint detection approach increased AU12's AUC from 0.67 to 0.74 and AU20's from 0.86 to 0.93. These results suggest that exploiting co-occurrence patterns can increase reliability – for instance, recognizing a Duchenne smile is easier when the model knows that AU6 and AU12 usually appear together. [Messinger et al., 2001] quite early demonstrated that different types of smiles in infants carry different emotional weights, with open-mouth cheek-raise smiles (involving AU6, AU12, and jaw drop) being most strongly associated with shared positive affect during face-to-face interactions. Most existing frameworks do not capture AUs in combination, instead treating each AU as an independent prediction or feature. Similarly, most affect classification studies typically use AUs as independent features without modeling their known relationships. Yet, there is evidence that configurations of AUs are more informative than any single AU in isolation. This insight motivates our attention to AU co-occurrences.

There has been a lot of work exploring deep learning methods for facial expression recognition. [Savchenko et al., 2022] proposed a lightweight framework designed for online learning environments. The models used are based on the EfficientNet-B0 architecture, pretrained on VGGFace2 dataset for face recognition, and then fine-tuned on the AffectNet dataset annotated with eight basic emotions (Anger, Contempt, Disgust, Fear, Happiness, Neutral, Sadness, and Surprise). Their approach focuses on achieving high accuracy maintaining low computational complexity. It achieved 62.4% accuracy on the Affect-

Net validation set. In our study, we leverage one of the models provided by EmotiEffLib and fine-tune it for infant affect recognition.

2 METHODS

2.1 SUPPORT VECTOR MACHINES (SVM)

Support Vector Machines (SVMs) are supervised learning classification models that are well regarded for their effectiveness in high-dimensional classification tasks with small to medium datasets. In our study, SVM serves as our primary classification approach for predicting affect from AU feature space. The goal of SVM is to find the optimal hyperplane that separates data points into different classes, maximizing the margin - the distance between the boundary and the nearest data points of each class - between them. The support vector machine (SVM) is an extension of the support vector classifier that results from enlarging the feature space in a specific way, using kernels [James et al., 2021].

In the binary classification setting, a linear SVM seeks a hyperplane that separates two classes and maximizes this margin. However, if data is not linearly separable, the SVM can be extended using kernel functions to project the original input features into a higher-dimensional space, where separation becomes feasible [Jakkula, 2006]. We adopt multiclass SVM [Weston and Watkins, 1999] to classify three classes in the feature space. The formulation comes down to:

$$\min_{w,b,\xi} \frac{1}{2} \sum_{m=1}^{3} \|w_m\|^2 + C \sum_{i=1}^{l} \sum_{m \neq y_i} \xi_{im} \tag{1}$$

$$\text{subject to} \quad w_{y_i}^\top x_i + b_{y_i} \geqslant w_m^\top x_i + b_m + 1 - \xi_{im}, \quad \xi_{im} \geqslant 0 \tag{2}$$

$$\hat{y} = \arg \max_{m \in \{1,2,3\}} \left(w_m^\top x + b_m \right) \tag{3}$$

Here:

- $x_i \in \mathbb{R}^d$ is the feature vector for sample i,
- $y_i \in \{1, 2, 3\}$ is its true class label,
- $w_m \in \mathbb{R}^d$ and $b_m \in \mathbb{R}$ define the hyperplane for class m,
- $\xi_{im} \geqslant 0$ are slack variables allowing margin violations,

- $C > 0$ trades off margin maximization against misclassification.

We used Radial Basis Function (RBF) kernel [Jakkula, 2006] to capture non-linear boundaries which is defined as:

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right) \tag{4}$$

where:

- x and x' are two input feature vectors,
- σ is a kernel parameter that controls the spread of the kernel function.

This formulation ensures that all data points are classified correctly, while simultaneously minimizing the norm of the weight vector. This leads to a decision boundary that generalizes well to unseen data, promoting both accuracy and robustness in classification.

2.2 TRANSFER LEARNING

As a complementary approach, we used transfer learning technique. It is a Deep Learning method in which instead of training a Neural Network from scratch, an already trained model from solving one problem (source task) is used to fine-tune it for another similar task with the final classification layer. Key benefits of this are that the model retains general feature representation by reusing a model's lower layers, adapting to the specific task and dataset. This is important, especially when the data available for the target task is limited [Yang and Zhang, 2020].

Transfer learning has proven to be highly effective across many domains, particularly in deep learning and computer vision.

A typical transfer learning workflow includes:

- **Pre-training**: A model is initially trained on a large dataset, typically from a general domain (e.g., BP4D+, which contains millions of labelled images).
- **Fine-tuning**: The pre-trained model is then adapted to the target task by replacing the final classification layers and retraining the model on the new dataset with a smaller learning rate. In some cases, only the top layers are retrained while the lower layers are frozen to preserve learned general features.

- **Model Evaluation**: After fine-tuning, the model is evaluated on the target task to assess its performance. Techniques like cross-validation are often used to ensure that the model generalizes well and does not overfit.

This method not only saves time and resources, but also improves the model's performance in situations where input data for the model is scarce or difficult to obtain. The ability to reuse pre-trained models makes transfer learning a powerful tool in both practical applications and research.

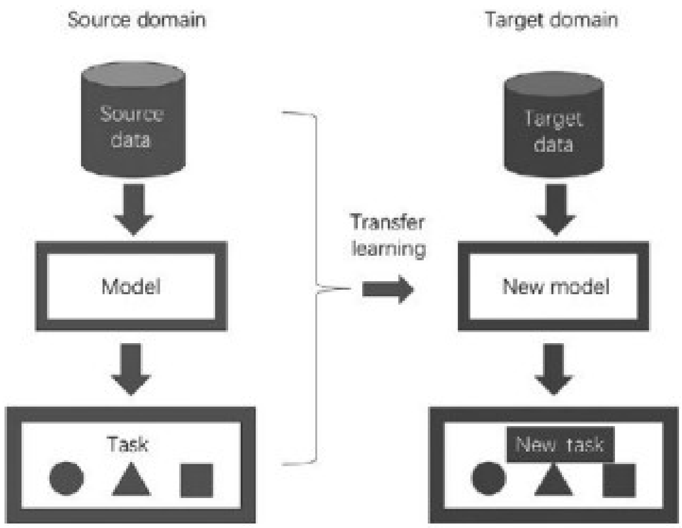

Figure 1: An illustration of the transfer learning process (from [Yang and Zhang, 2020]). The figure shows how a model pre-trained on a source task is adapted to a new target task with new data.

2.3 CONVOLUTIONAL NEURAL NETWORKS (CNN)

Convolutional Neural Networks (CNNs) are deep learning models designed to automatically learn hierarchical features from raw input data, particularly image data. CNNs are composed of several layers, including convolutional layers, pooling layers, and fully connected layers [James et al., 2021].

- Convolutional Layers: These layers apply filters (kernels) to the input image, extracting local features such as edges, textures, and shapes. Each filter detects different patterns in the image.

- Pooling Layers: Pooling layers downsample the feature maps, reducing the dimensionality and retaining the most important information. This makes the model more computationally efficient.
- Fully Connected Layers: These layers are used for classification after feature extraction. The output of the convolutional and pooling layers is flattened into a one-dimensional vector, which is passed through one or more fully connected layers to produce the final class probabilities.

CNNs can be particularly effective for tasks like ours such as affect recognition or identification, where facial features need to be learned automatically from images.

2.3.1 EfficientNet B0 architecture

CNNs can be scaled up or down by adjusting network layers, they can also be scaled in width to achieve better accuracy. Bigger images as input can also help with improving accuracy. [Tan and Le, 2019] have provided a new way of scaling up that scales all three of these parameters with a fixed ratio. EfficientNet-B0 architecture is part of the EfficientNet family, a set of models introduced by [Tan and Le, 2019] to address the problem of scaling CNNs efficiently. EfficientNet-B0 is the baseline model in the EfficientNet series and has been designed to achieve state-of-the-art performance while maintaining computational efficiency. The core innovation of EfficientNet lies in its compound scaling method, which uniformly scales the network's depth, width, and resolution using a fixed set of scaling coefficients.

Unlike conventional models, where scaling typically involves arbitrarily increasing depth, width, or image resolution independently, EfficientNet-B0 scales all three dimensions simultaneously in a principled manner. This scaling approach is based on a grid search to find the optimal scaling coefficients for depth, width, and resolution, ensuring a balanced growth of the network. This results in a network that is not only more efficient but also provides better performance with fewer resources.

The architecture of EfficientNet-B0 includes:

- **Mobile Inverted Bottleneck Convolutions**: EfficientNet-B0 uses MBConv blocks, which are highly efficient and designed for mobile applications. These blocks include depthwise separable convolutions, which reduce the number of parameters and computations without losing much in performance.

- **Squeeze-and-Excitation (SE) Blocks**: The SE block is used to recalibrate channel-wise feature responses, helping the model focus on more informative features.
- **Final Layer**: The network ends with a global average pooling layer followed by a fully connected layer for classification.

EfficientNet-B0 performs better than other models such as ResNet [He et al., 2016] and DenseNet[Huang et al., 2017] in terms of accuracy while being much smaller and faster. This model can be effective for transfer learning, as it allows for fast fine-tuning on smaller, task-specific datasets. It achieves 77.1% top-1 accuracy on ImageNet, outperforming other models with fewer parameters. ImageNet is a benchmark often used to evaluate the performance of image classification models [Deng et al., 2009].

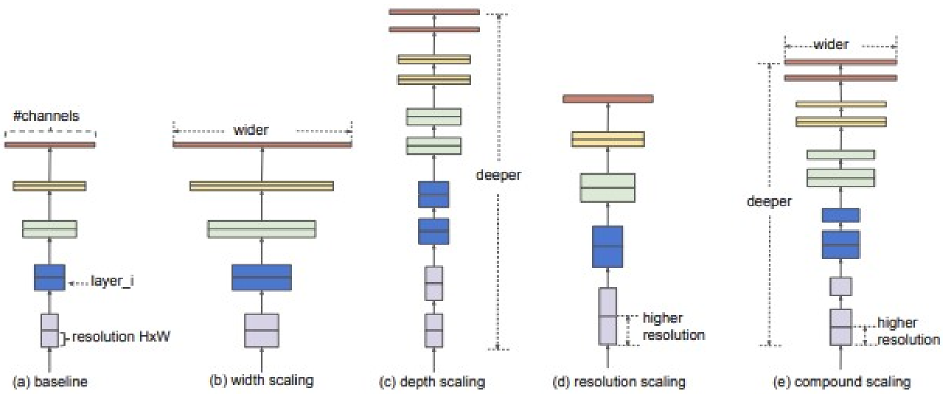

Figure 2: An illustration of the Model Scaling process (from [Tan and Le, 2019]) (a) is a baseline network example; (b)-(d) are conventional scaling that only increases one dimension of network width, depth, or resolution. (e) is [Tan and Le, 2019] proposed compound scaling method that uniformly scales all three dimensions with a fixed ratio.

The compound scaling method is an advancement in the field of CNN design and has proven to be highly effective in various tasks, including image recognition tasks. The ability to scale the model efficiently ensures that it achieves good accuracy while requiring less resources, which is important for practical applications.

2.4 SALIENCY MAPS AND SMOOTHGRAD

Interpreting the inner workings of neural networks is essential for understanding model predictions. This section draws primarily on the interpretability framework described by [Molnar, 2022].

A commonly used techninque for interpreting image-based models is the saliency map, also known as pixel attribution. As explained by Molnar, it identifies the pixels in an image that contribute most to the model's prediction by computing the gradient of the output with respect to the input image. That is represented by the gradient $\nabla_x f(x)$ that tells us how changes in each pixel affect the output, highlighting the important features in the image.

However, raw saliency maps can suffer from high-frequency noise - spiky, unstructured pixel importance. SmoothGrad, initially introduced by [Smilkov et al., 2017] improves upon saliency maps by reducing the noise in the gradient visualizations. It is a simple method that can help visually sharpen gradient-based sensitivity maps. It works by adding Gaussian noise to the input image multiple times, generating several versions, and then averaging the gradients across these noisy images. This process smoothens out random fluctuations, resulting in clearer and more stable saliency maps that reveal which parts of the image influence the model's prediction the most [Molnar, 2022]. While saliency maps provide insight into the model's behavior, using SmoothGrad technique for saliency maps produces cleaner and more reliable attributions.

3 LIBRARIES AND DATASETS

3.1 CITY INFANT FACES DATASET

The dataset used in our research is The City Infant Faces database. It is a validated set of 154 portrait images capturing infant facial expressions, of 69 different infants. These images represent infants aged 0–12 months and are categorized into three affective states: 60 positive, 54 negative, and 40 neutral expressions. Positive expressions include smiling, laughing, and excited faces, while negative expressions include sadness, anger, worry, fear, and distress. [Webb et al., 2017]

Figure 1: Example images from the City Infant Faces dataset showing the same infant with expressions for the three affective categories: positive (B5Pos), neutral (B5Neu), and negative (B5Neg).

The database was developed in three stages: collection, validation, and reliability testing. Parents were recruited via social media platforms and asked to provide images of their infants showing a minimum of three photographs of their infant showing at least one positive, one negative, and a neutral emotion. A total of 195 images were collected from 68 parents. All images were processed in Adobe Photoshop in 2014, standardized to a resolution of 800 × 1,100 pixels, had their backgrounds removed and presented in both black-and-white and color formats (only black-and-white images were fully validated).

The validation process involved 71 participants — 41 midwifery students, 12 neonatal nursing students, and 18 members of the general public with a mean age of 28 years, who rated each image on six dimensions: expression (positive/neutral/negative), intensity (how strongly the participants felt the face was showing this emotion), clarity (how clear the expression was), genuineness (how genuine the participant felt the expression was), affective response (what emotion the participant felt while viewing the image), and strength of affective response (how strongly the participant felt this emotion). Ratings were collected using PowerPoint presentations and online surveys. Only images with at least 75% inter-rater agreement on expression classification were retained in the final database — 41 images were removed from the database, with rejected images having poorer agreement that ranged from 46.5% to 70%. This ensured a high agreement rate of 91.76% overall, with 95.73% agreement for positive, 94.87% for negative, and 84.7% for neutral expressions.

Additionally, test–retest reliability was assessed by re-rating a subset of images four weeks later with midwifery students, providing excellent correlation scores for negative (r = .954) and neutral (r = .965) expressions, and good reliability for positive expressions (r = .655). These results suggest that the City Infant Faces database provides a naturalistic yet controlled set of infant emotional expressions suitable for affect recognition research.

3.2 PYAFAR: AUTOMATED FACIAL ACTION RECOGNITION LIBRARY

PyAFAR is a Python-based, open-source facial action unit detection library for use with adults and infants. It has gone through three different phases of architecture changes and adaptations to AU extraction approaches [Hinduja et al., 2023], [Onal Ertugrul et al., 2023], we focus on the latest enhanced version [Ertugrul et al., 2024]. It combines real-time facial landmark tracking, face normalization, and AU prediction in a unified framework. PyAFAR has two distinct AU detection pipelines, one previously mentioned as PyAFAR INFANT, the other - PyAFAR ADULT, each with a different architecture and training approach.

PyAFAR ADULT is based on a deep learning architecture that was trained on BP4D+ dataset [Zhang et al., 2016], described in next subsection. The model predicts the occurence of 12 AUs nad estimates intensity for five of these. Facial landmark detection and tracking of 3D landmarks is performed using MediaPipe [Lugaresi et al., 2019]. These landmarks are then used to normalize the face via transformation, aligning it to a canonical pose using dlib library [King, 2009]. The normalized face image is then passed to a CNN. These networks operate in a one-versus-all configuration and use softmax or sigmoid activations depending on whether intensity or binary occurrence is predicted. The outputs are occurance values or intensity scores for each frame (for videos) or for each image, which are stored in CSV and JSON formats. The adult model is not able to detect AU3, AU9, AU20 and AU28 - AUs that PyAFAR INFANT captures, but will detect AU7 (lids tight), AU10 (upper lip raiser), AU14 (dimpler), AU15 (lip corner depressor), AU17 (chin raiser), AU23 (lip tightener) and AU24 (lip presser). However, the AUs only captured by PyAFAR ADULT model were not used in our further analysis.

PyAFAR INFANT [Ertugrul et al., 2024] has a fundamentally different architecture that prioritizes computationnal efficiency. Rather than using deep learning, this model extracts Histogram of Oriented Gradients (HoG) features [Déniz et al., 2011] from aligned face images, which are resized to 112×112 pixels. HoG features are extracted using 2×2 cells per block, each having 8×8 pixels per cell and 9 orientations resulting in 6084 dimensional vectors. These features are used to train individual light Gradient Boosting Machines (lightGBM) classifiers per AU that output occurence probabilities of AUs similarly like PyAFAR ADULT does - for each frame (for videos) or for each image.. HoG features capture the distribution of edge orientations in localized portions of an image. LightGBM is a decision tree ensemble method that constructs trees in a leaf-wise manner, enabling it to focus on regions of the feature space with the greatest classification error. The HoG features improve computational efficiency by running a similar task as PyAFAR ADULT more tha $5\times$ faster. In infants, the AU occurrence can be predicted for 9 Action Units: AU1 (inner brow raiser), AU2 (outer brow raiser), AU3 (inner brows drawn together), AU4 (brow lowerer), AU6 (cheek raiser), AU9 (upper lip raiser), AU12 (lip corner puller), AU20 (lip stretcher) and AU28 (lip suck). These actions units are represented in Figure 2.

Figure 2: Action Units automatically detected with PyAFAR INFANT model, from [Onal Ertugrul et al., 2023].

3.2.1 Datasets for PyAFAR: MIAMI & CLOCK

The PyAFAR ADULT is trained on BP4D+ [Zhang et al., 2016], a multimodal spontaneous emotion corpus that includes synchronized 3D, 2D, thermal, and physiological data sequences, providing annotated videos of adult facial expressions during a range of tasks. In the dataset there are 140 subjects - 58 males and 82 females in the age range of 18-66 years old. Ethnic/Racial Ancestries include Black, White, Asian (including East-Asian and Middle-East-Asian), Hispanic/Latino, and others (e.g., Native American). The dataset contains 10 tasks (emotions) for each subject, there are over 10TB of high quality data generated for the research community. Only action units with a base rate greater than 5% in the training data are included, ensuring reliable detection.

PyAFAR INFANT leverages two well-annotated datasets. MIAMI [Hammal et al., 2015a], which contains spontaneous behavior data from 42 ethnically diverse 4-month-old infants and their mothers. Infants were recorded with their mothers in a Face-to-Face/Still-Face (FF/SF) protocol [Adamson and Frick, 2003] that elicits both positive and negative affect. FF/SF protocol assesses infant responses to parent unresponsiveness. It has three episodes: parent and infant engage in face-to-face intraction (FF), the parent stops interacting with the infant and gazes at them with a neutral expression (SF), and the parent-infant interaction resumes.

And CLOCK (Craniofacial microsomia: Longitudinal Outcomes in Children pre-Kindergarten) [Bian et al., 2023], which provides additional infant facial expression data from infants with craniofacial microsomia (CFM) aged 12 to 24 months that were followed to age 36 to 48 months. 44 cases ad 36 controls were observed and video recorded. Specifically, two age-appropriate emotion induction tasks were used to elicit spontaneous positive and negative facial expressions. In the positive emotion task, an experimenter blew soap bubbles towards the infant. In the negative emotion task, an experimenter presented a toy car to the infant, allowed the child to touch it, then retrieved the car and covered it with a transparent plastic bin.

3.3 EMOTIEFFLIB: LIBRARY FOR EFFICIENT EMOTION RECOGNITION

EmotiEffLib is a lightweight, open-source library for emotion and engagement recognition in photos and videos [Savchenko et al., 2022]. It was developed as a fast and accurate technique that can be implemented in online learning software at laptops without powerful GPUs, for environments with constrained computational resources. The result is lightweight FER (facial expression recognition) models based on EfficientNet architecture (Section 2.3.1), that supports fast inference while maintaining high accuracy. They reached the state-of-the-art results in several emotion recognition tasks.

EmotiEffLib provides several models and customizable emotion, affect and engagement prediction pipelines, allowing for adaptations to specific tasks. The library offers models like MobileNet-v1 for resource-constrained environments, EfficientNet-B0 for balanced accuracy and efficiency, and EfficientNet-B2 for higher accuracy in more demanding applications. This flexibility includes the ability to fine-tune models or use different datasets for training based on concrete applications.

The core model we used in our study is based on the EfficientNet-B0 architecture. It is capable of classifying basic facial emotions (Anger, Contempt, Disgust, Fear, Happiness, Neutral, Sadness and Surprise), predicting engagement levels (e.g., from disengaged to highly engaged), and handling affective states. The overall accuracy compared to other models provided from the EfficientNet family is slightly lower, but still comparable with the best-known results reported for AffectNet dataset [Tao and Tan, 2005], with 8-class accuracy 61.32 vs 63.03 for EfficientNet-B2. It is important to emphasize that though the average accuracy of the deepest EfficientNet-B2 model is greater, the class accuracy for every type of emotions is sometimes lower when compared to EfficientNet-B0 [Savchenko et al., 2022]. We chose EfficientNet-B0 model because of balanced inference speed and efficiency.

It was initially trained on the VGGFace2 dataset [Cao et al., 2018] to learn a broad range of facial features that can later be adapted. On the second training stage, the model was fine-tuned on the AffectNet dataset [Tao and Tan, 2005] for emotion and affect recognition. It was trained on an imbalanced subset of 287,651 images annotated with 8 categorical emotion labels. Finally, a subset of 4000 images (500 per class) was used for validation. Though the AffectNet contains mainly photos of adults, the resulting FER models can classify emotions even for young children [Savchenko et al., 2022]. The model was trained on 8 epochs, with Adam-based optimizer and learning rate of 0.001. The training happened on cropped facial images, ensuring that the model focuses on facial features without the influence of background elements. Pre-processing like face alignment were applied. The source for training and testing code is the Tensorflow 2 and Pytorch framework.

3.3.1 Datasets used for training EmotiEffLib model: VGGFace2 & AffectNet

Initially, the model `enet_b0_8_best_afew` using EfficientNet-B0 architecture was pre-trained on VGGFace2 [Cao et al., 2018], a dataset consisting of over 3 million images of 9131 subjects, for face recognition. It consists of images from Google Image Search, and have large variations in pose, age, illumination, ethnicity and profession. The pre-training captured general facial features.

AffectNet [Mollahosseini et al., 2017] is a large-scale facial expression dataset containing over 1 million facial images and extracted facial landmark points from the internet by querying different search engines (Google, Bing, and Yahoo) using emotion related tags in six different languages (English, Spanish, Portuguese, German, Arabic, and Farsi). Twelve human experts manually annotated 45000 of these images in both categorical and dimensional (valence and arousal) models. For categorical model, labellers used the following labels: neutral, happy, sad, surprise, fear, anger, disgust, and contempt. The dataset is highly diverse, with images representing a wide range of ethnicities, ages, and real-world conditions like variations in lightning, poses and occlusion (when two objects are too close to each other). It is important to note that AffectNet's annotations can be noisy as human annotators agreed on 60.7 percent of the category of facial expressions, proving that expression recognition and predicting valence and arousal in the wild is a challenging task [Mollahosseini et al., 2017]. Though, AffectNet is by far the largest database of facial affect in still images which covers both categorical and dimensional models.

4 EXPERIMENTS

In this section, we present the experimental settings for assessing the effectiveness of our infant affect recognition models. We investigate two approaches for classifying affect: the first one is a Support Vector Machine (SVM) model using the AUs extracted by PyAFAR INFANT as features, and the second is a deep learning approach based on transfer learning with a pre-trained CNN model from EmotiEffLib.

4.1 SVM CLASSIFICATION

A Support Vector Machine (SVM) model was chosen to classify the images into positive, negative and neutral affect categories. For selecting the hyperparameters, we performed cross-validation with 5 folds. The final SVM model uses the Radial Basis Kernel (RBF), the regularization parameter 'C' of 10 and gamma = 'auto'.

The baseline model included all 9 Action Units extracted using PyAFAR INFANT. The evaluation metric used for assessing the model's performance is the class accuracy. It measures the proportion of correctly classified samples within each affect category. To better understand the model's limitations, misclassified images were further examined to investigate potential patterns or factors contributing to incorrect predictions.

Due to the small size of the data set, we used the Leave-One-Out (LOO) cross-validation method to evaluate the performance of the final model. This method trains the model iteratively on the entire dataset but one sample is used to test the trained model. This is done repeatedly for each image in the dataset, so that each one is used for validation exactly once, allowing a less biased evaluation.

We also trained two SVM models using features extracted by PyAFAR INFANT and PyAFAR ADULT, respectively. For a fair comparison, we only used as features the five AUs that are common across both models: (AU1, AU2, AU4, AU6, AU12).

4.2 FINE-TUNING CNN MODEL

We performed transfer learning using the `enet_b0_8_best_afew` model, as described in Section 3.3. For our task, the images in our dataset were resized to 224x224 pixels and normalized accordingly, to ensure consistency with the model. The feature extraction layers of the pre-trained model remained frozen, while the last layer was replaced with a fully connected layer with 3 output neurons and Softmax as final activation function for classification between affect labels.

The validation technique for this approach was a 5-fold cross validation. The dataset was initially divided into 5 different splits with proportions 80-20% for train and test sets respectively. For each fold, it was ensured that the same infant id was not used in both train and test together, and that every single image was used exactly once at any of the folds. A 5-fold cross validation offers a better generalization compared to a single random split because it ensures the model is trained and evaluated on different subsets of the data, reducing the risk of overfitting to a specific train-test division.

For each of the folds a model was trained with 8 epochs and a batch size of 8, using Cross-Entropy Loss and Adam optimizer with learning rate of 0.001. The original choice for this parameters followed the same configuration as in the original EmotiEffLib implementation. As shown in Figure 1, the training loss consistently decreased over the 8 epochs, with a rate of improvement slowing after epoch 6. Further training may have offered gains, but for computational efficiency we stopped at 8 epochs.

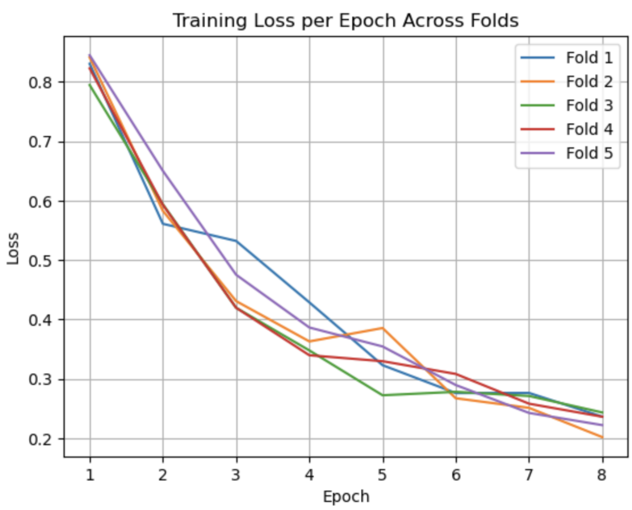

Figure 1: Training loss vs epoch across the 5 cross-validation folds.

The performance was measured again with accuracy metric on the test size. The final reported accuracy is defined as the average of the accuracy scores across the 5 folds.

To validate that the model is actually based on facial expressions rather than background artifacts of the images, we applied Smooth-Grad method with number of noisy samples of 100 and 0.05 of noise standard deviation.

5 RESULTS

In this section, we present the results of our experiments assessing the use of PyAFAR INFANT tool for affect classification. We begin by analyzing the AUs detected by the model and their relationship with different classes. We then evaluated the performance of a SVM classifier that uses these AUs as features to classify affect, and investigate the most frequent misclassifications types. We compare this model with another SVM that use AUs extracted with the PyAFAR ADULT model, and we conclude with a deep-learning approach, fine-tuning a CNN pre-trained on emotion recognition. The results across models are finally compared in the last section.

5.1 ANALYSIS OF ACTION UNITS DETECTED BY PYAFAR INFANT

To evaluate the effectiveness of the PyAFAR tool for predicting affect, we started with a manual inspection of the AUs activation probabilities estimated by the INFANT model on a subset of the dataset. Figure 1 shows six different images, and the corresponding AU occurrence probabilities.

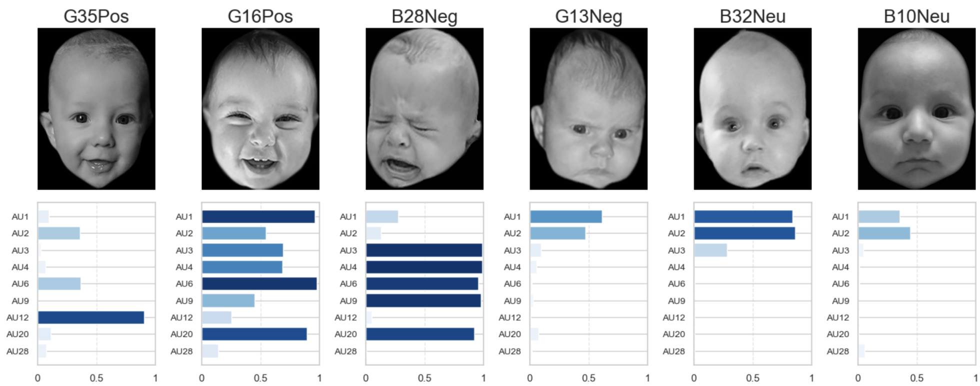

Figure 1: Subset of images and corresponding probability of occurrence of each of the AUs, estimated by PyAFAR INFANT.

In the first positive sample (G35Pos), the model show to correctly identify a smile with a high probability on AU12 (lip corner puller) and a moderate value of AU6 (cheek raiser), consistent with a subtle eye constriction. This matches the expectation for smiles, defined in the literature (Table 1, Section 1). However, in the second smiling infant (G16Pos), AU6 and AU9 (upper lip raiser or nose wrinkle) are properly detected, but AU12 is not activated. It is AU20 (lip stretcher) that show high value, which indicates that there is a confusion between smiling and horizontal mouth stretching, given that we can see a clear smile in the image.

In the first negative image (B28Neg), the model detected with high probability the activation of AU20, meaning that is accurately detecting the lip stretch that is associated with the crying face, clear in the image. This also matches the expected. Other AUs as AU3 and AU4 (brow lowerer), AU6 and AU9, were also correctly detected with high probabilities. However, in the second negative image (G13Neg), the model seems to miss the detection of AU3 and AU4, when we can clearly see a furrowed brow. This indicates that the model might struggle detecting brow-related AUs as well.

For neutral images (B32Neu and B10Neu), the model accurately detect moderate values of the brow raisers AU1 and AU2, and low probability on the rest of the AUs.

From this initial manual inspection, we can say that the PyAFAR INFANT model generally performs well, but it may confuse AU12 (smiles) with AU20 (lip stretch), and could under-detect brow-related AUs, particularly AU3 (inner brows drawn together) and AU4 (brow lowerer).

5.1.1 Mean AU Occurrence per Affect Class

To further evaluate the effectiveness of the PyAFAR INFANT, we computed the mean probability of occurrence of selected AUs across the three affect classes: positive, negative, and neutral. This serves as a comparison point to the previous work of [Zaharieva et al., 2023]. who used Baby FaceReader 9 for AU extraction.

As shown in Figure 2, AU6 (cheek raiser) and AU12 (lip corner puller) had the highest mean activation on positive affect, while AU3+AU4 (obtanined as an average of individual AUs) and AU20 are more prominent in negative expressions. This aligns with the finding of [Zaharieva et al., 2023]. AU1 and AU2 (brow raisers) appear more frequently on neutral faces. Neutral expressions in general showed very low probabilities of occurrence of AU, suggesting minimal facial movement when infants are in a neutral state.

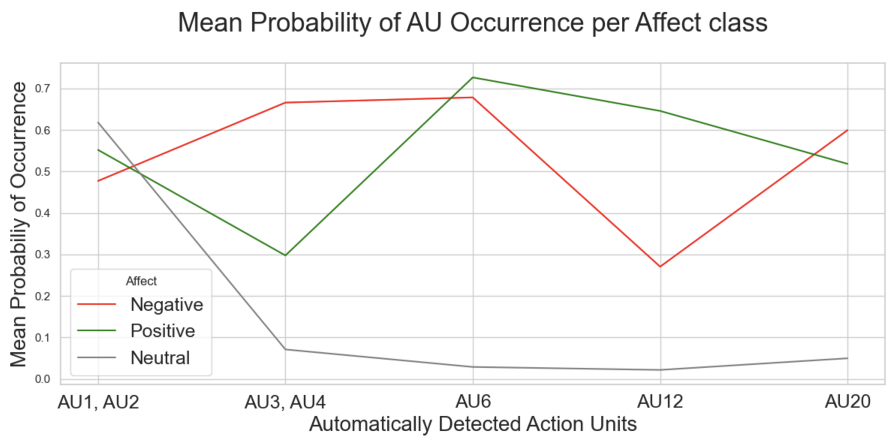

Figure 2: Mean probability of occurrence of AUs detected by PyAFAR INFANT across positive, negative, and neutral affect classes.

We therefore defined a baseline expectation of typical AU occurrence patterns for each of the affect classes. The results are displayed in the bar plots in Figure 3. In this figure, each bar represents the average probability of occurrence of each AU for a given affect class.

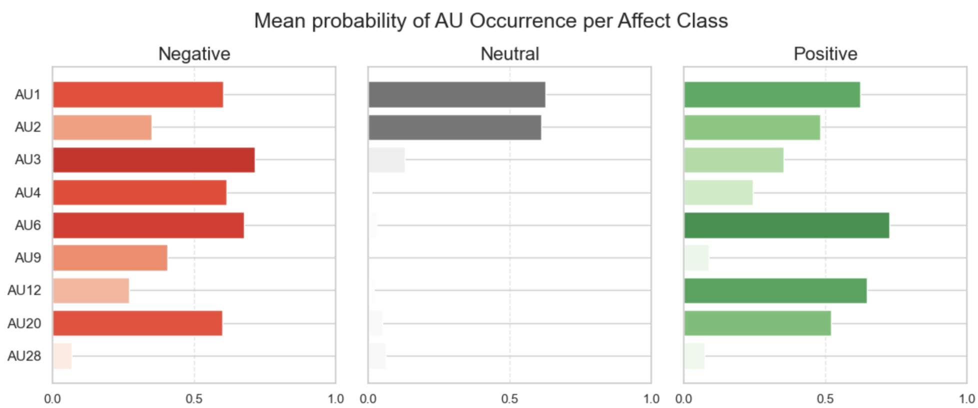

Figure 3: Mean probability of occurrence of Action Units (AUs) for each affect class, detected by PyAFAR INFANT. Baseline profiles for AU patterns associated with positive, negative, and neutral infant affect.

Despite some detection errors identified with manual inspection (Figure 1), the findings after analyzing the entire dataset are consistent with previous work, and we conclude that PyAFAR INFANT can accurately capture typical AU patterns associated with infant affect. However, confusion between AU12 (smile) and AU20 (lip stretch), and under-detection of brow-related AUs, could be be a limitation of the model.

5.2 AUS DETECTED BY PYAFAR ADULTS

We compared AU detection results between the PyAFAR INFANT and ADULT models using the five AUs common on both models: AU1, AU2, AU4, AU6, AU12. Figure 4 shows the same sample set analyzed in the previous section, now with the AU probabilities estimated by both models. We can see that in most of the cases the adult model fails to detect the brow-related AUs (AU1, AU2, AU4), returning near zero probabilities. This was expected, due to morphological differences in infant facial structures, particularly in the brow area.

Moreover, we can observe differences in the detection of AU12. The adult model incorrectly detects a smile in cases like B28Neg that has a clear negative expression, while it correctly detects the smile on G16Pos, a case that the infant model failed to detect.

These findings reinforce the importance of using AU detectors specifically trained on infants.

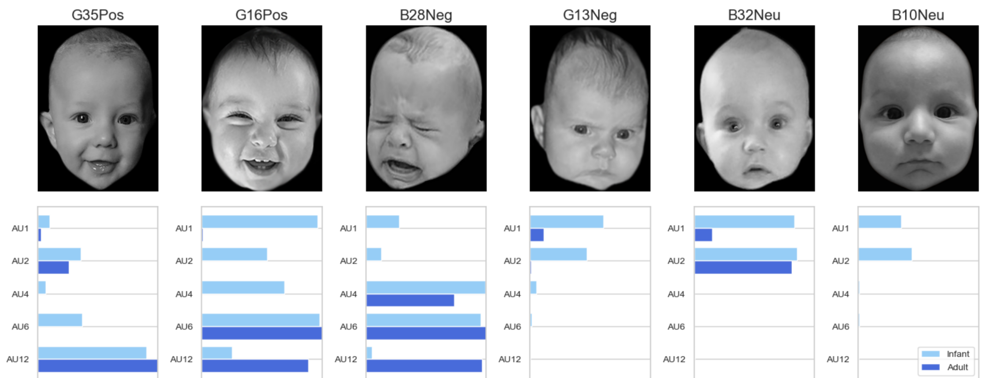

Figure 4: Comparison of AU probabilities between PyAFAR INFANT and ADULT. Adult model show limited detection of brow-related AUs.

5.3 SVM CLASSIFICATION PERFORMANCE

After analyzing AUs estimates across affect classes, a Support Vector Machine (SVM) classifier was trained to predict infant affect, using all PyAFAR INFANT-extracted AUs as features.

Table 1 shows the resulting classification accuracy, overall and class-specific. The model achieved an overall accuracy of 0.86. Positive and neutral classes were predicted with high accuracy (0.92 each), while the negative class had the lowest score (0.76), suggesting some difficulty in distinguishing infant negative affect.

Table 1: SVM classification accuracy per affect class

Affect Class	Negative	Neutral	Positive	Overall
Accuracy	0.76	0.92	0.92	0.86

Figure 5 shows the confusion matrix for the SVM classifier. Most of the misclassifications occurred in the negative affect class, with 7 negative samples incorrectly predicted as positive, and 5 as neutral. These misclassified images are displayed and further analyzed in the next section.

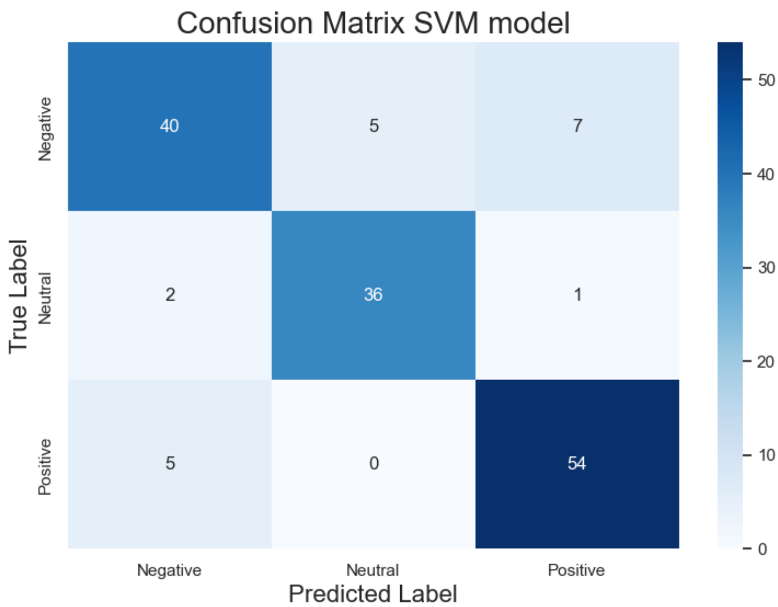

Figure 5: Confusion matrix of the SVM classifier with all AUs as features.

5.4 ANALYSIS OF MISCLASSIFICATIONS

In the previous section, we observed that the SVM classifier performed significantly worse on the negative class, with 0.76 class-accuracy compared to 0.92 for both positive and negative. To better understand this discrepancy and limitations of the model, we do a qualitative analysis of the most frequent misclassification types. This way we can provide an insight into which Action Unit (AU) patterns are associated with incorrect predictions.

5.4.1 Negative Samples Misclassified as Neutral

The Figure 6 displays all five negative samples that were incorrectly classified as neutral affect, with their respective AU activation profile.

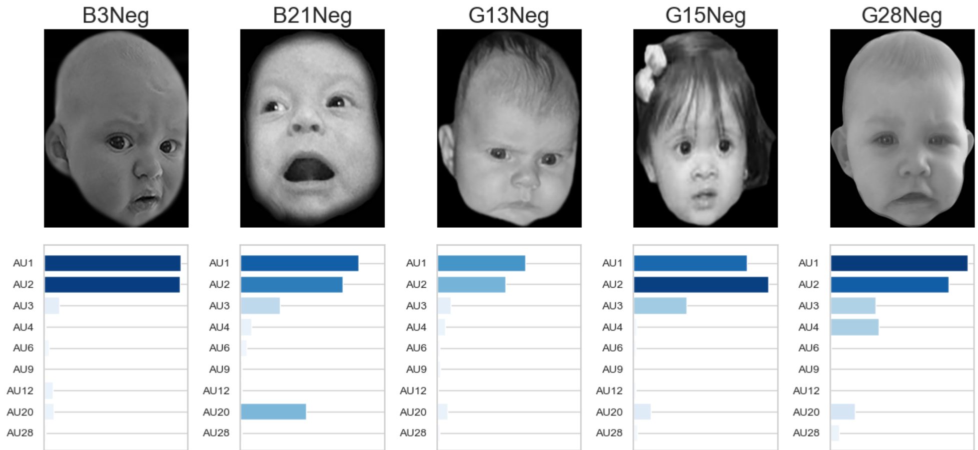

Figure 6: Negative images misclassified as NEUTRAL and respective AU activation values.

The five misclassified images consistently show weak activation in AUs that are typically strong indicators of negative affect, based on the baseline defined in Figure 3: AU3 and AU4 (brow lowerer) and AU20 (lip stretcher). On the other hand, we observe very high activation of AU1 and AU2 (brow raiser), while the remaining AUs show very low activation. This aligns with the neutral activation profile in Figure 3, the main reason behind these misclassifications.

The boxplots in Figure 7 reinforce this observation. The misclassified samples, represented by colored dots, consistently exhibit higher activation of AU1 and AU2, and lower activation in the rest of the AUs, compared to the typical distribution for the negative class.

In samples B3Neg, G13Neg and G15Neg, the incorrect prediction is explained by a lack of detectable activation in AU3 and AU4. This incapability was already noticed in the previous section. In other cases, like image G28Neg, the negative expression itself is subtle. Even with an accurate AU detection, the misclassification can be explained by the ambiguity in infant facial expressions, which could make it difficult to distinguish subjectively also for human observers.

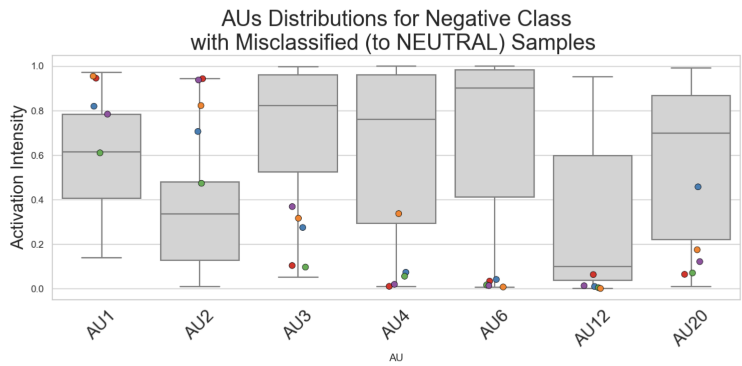

Figure 7: Distribution of AU activation for negative samples. Dots represents negative samples misclassified as neutral.

5.4.2 Negative Samples Misclassified as Positive

The seven negative samples that were misclassified as positive are shown in Figure 8, together with the corresponding AU activation profiles.

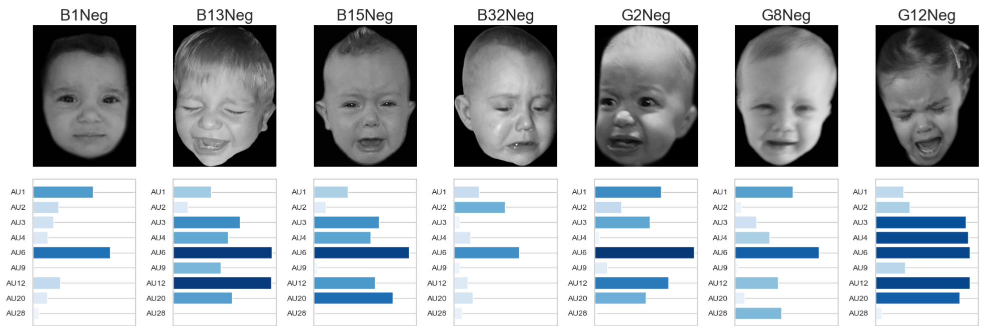

Figure 8: Negative images misclassified as POSITIVE and respective AU activation profile.

Across these samples, we observe a pattern: most images show a strong activation of AU6 (cheek raiser) and AU12 (lip corner puller), which are typically associated with smiling and mostly activated in positive labeled images (Figure 3). However, these images also present high activation of AU3, AU4 and AU20, AUs typically linked to negative affect.

This suggests that the classifer is defaulting to predict positive affect when AU6 and AU12 are strongly activated, even when negative cues (AU3, AU4 and AU20) are simultaneously present. This is a particular challenge in infant affect recognition, as infants often present complex expressions that involve co-activation of both positive and negative linked AUs.

5.4.3 Positive Samples Misclassified as Negative

Figure 9 show the five positive samples incorrectly classified as negative. A common pattern shared across these images is the presence of eye constriction, and low visibility of the brows. These characteristics appear to result in an incorrect high activation of AU3 and AU4. Aditionally, AU20 (lip stretcher) is activated instead of AU12, on cases where there are clear smiles.

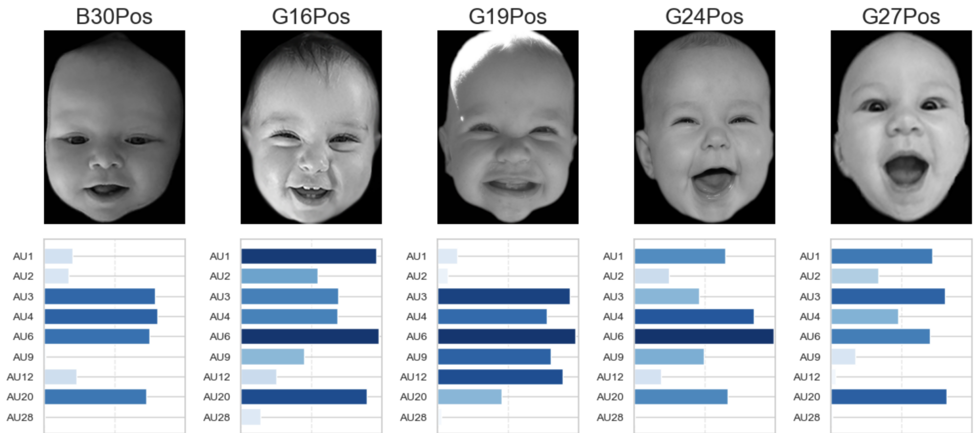

Figure 9: Positive images misclassified as NEGATIVE and respective AU activation profile.

These findings show that relying only on individual AU activations detected by PyAFAR INFANT may not be enough to predict infant affect, especially in more subtle or overlapping expressions. These observations motivate us to explore if a deep learning approach can better handle these complex AU combinations, to address ambiguities in infant expressions.

5.5 PYAFAR ADULT: BENCHMARKING ON INFANT DATA

A final comparison was made between two SVM models, using as features AUs extracted by two different models: PyAFAR INFANT and PyAFAR ADULT. The only AUs that are common on both models are AU1 (inner brow raiser), AU2 (outer brow raiser), AU4 (brow lowerer), AU6 (cheek raiser), AU12 (lip corner puller), so we trained both SVM models only on those 5 features for a fair comparison.

The overall and class accuracies are shown in Table 2. The infant model achieved an overall accuracy of 0.83, outperforming the

Table 2: SVM classification accuracy per affect class using PyAFAR INFANT vs ADULT model for feature extraction, using only common Action Units (AU1, AU2, AU4, AU6, AU12).

PyAFAR Model	Negative	Neutral	Positive	Overall
INFANT	0.71	0.97	0.83	0.83
ADULT	0.53	1.00	0.80	0.75

adult model, with 0.75 accuracy. The biggest difference is in the Negative class, in which the Adult model had a class accuracy of 0.53, just above chance. The reason might be that negative affect is often expressed through brow lowering (AU3 and AU4), which is morphologically very different from adults to infants. A model trained only on adults may struggle to correctly detect this action units on infants. Moreover, AU20 (lip stretcher), which is characteristic of negative affect, is not present in the AUs extracted by the Adult model.

These findings confirm the idea that models trained on adult facial expressions fail to generalize to infants.

5.6 TRANSFER LEARNING WITH CNN: FINE-TUNING EMOTIEFFLIB MODEL

After evaluating an AU-based classifier, we explored a deep learning approach by fine-tuning the model provided by EmottiEfflib, originally trained on the AffectNet dataset for emotion recognition.

We trained and tested the model on the five different splits of our dataset, previously explained in Section 4.2. The final overall and class accuracies are presented in Table 3. The model achieved an overall accuracy of 0.96, significantly outperforming the previous SVM model.

Table 3: Fine-tuned CNN model accuracy per affect class

Affect Class	Negative	Neutral	Positive	Overall
Accuracy	0.92	1.00	0.97	0.96

The CNN model presents slightly lower accuracy for the negative class (0.92) compared to neutral (1.00) and positive (0.97). However, the difference is smaller than the gap observed with the SVM classifier. The confusion matrix in Figure 10 illustrates how misclassifications are distributed across classes. The number of negative images misclassified as positive has significantly been reduced, which was the major issue in the previous SVM model. This improvement indicates that the CNN model has a greater capacity to distinguish between positive and negative affect.

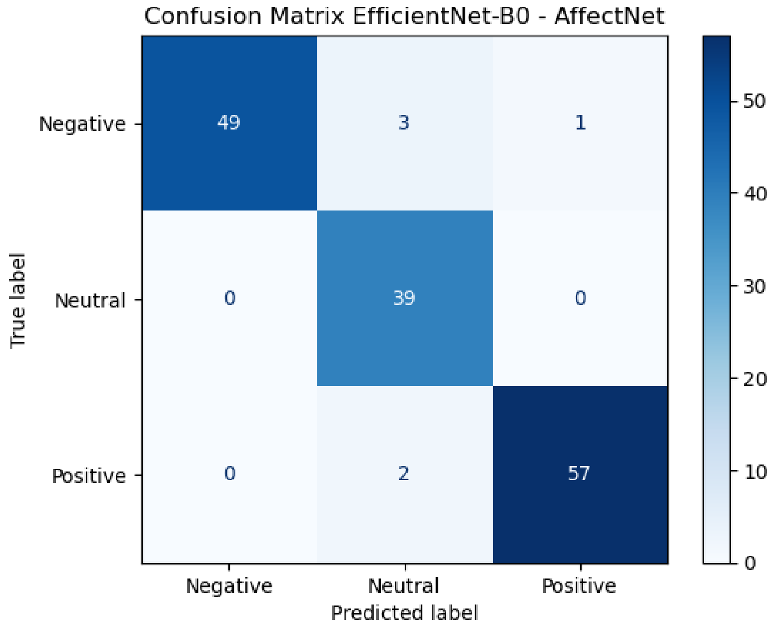

Figure 10: Confusion matrix of the fine-tuned CNN classifier.

To validate the CNN model, we generated SmoothGrad saliency maps. These maps show which regions of each image contributed most strongly to the decision of the model.

Figure 11 shows the heatmaps for the seven misclassified images of the model. In all of them, the highlighted areas corresponds to the main expressive facial regions, like eyebrows, eyes and mouth. This confirms that the model is indeed learning from facial features, responsible for the decision making of the classification, rather than making random predictions or based on background features.

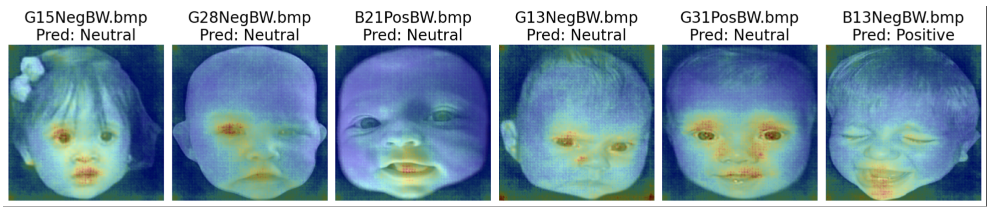

Figure 11: Saliency maps for misclassified images by the CNN fine-tuned model.

The analysis of the misclassifications in Figure 11 shows that, similar to the SVM model, the CNN struggles particularly with the same negative images that are incorrectly classified as neutral: G15Neg, G28Neg and G13Neg. However, the negative expressions in these images are very subtle, making the affect state difficult to distinguish. This suggests that the misclassifications are explained again due to the ambiguity of the facial expression itself.

Compared to the previous SVM classifier, the CNN shows a substantial improvement, particularly in the negative vs positive classifications. This confirms the greater capacity of the deep learning model to capture more complex facial expressions.

5.7 COMPARATIVE EVALUATION: AU-BASED VS CNN-BASED MODELS

We conclude the results section with a comparison of the performance of the four different models implemented in this study. The first three are Support Vector Machine (SVM) classifiers using as features the AUs detected by PyAFAR INFANT or PyAFAR ADULTS. These differ in their input features: the first model uses the full set of 9 AUs available from PyAFAR INFANT (Section 5.3), while the second and third use only the 5 AUs that are common between PyAFAR INFANT and PyAFAR ADULT (Section 5.5). The final model is a fine-tuned Convolutional Neutral Network (CNN) pre-trained on AffectNet for emotion recognition (Section 5.6).

Table 4 displays the overall and class-accuracies across models.

Table 4: Comparison of Affect Classification Accuracy Across Models

Model	Neg.	Neu.	Pos.	Overall
SVM - INFANT full AUs	0.76	0.92	0.92	0.86
SVM - INFANT common AUs	0.71	0.97	0.83	0.83
SVM - ADULT common AUs	0.53	1.00	0.80	0.75
Fine-tuned CNN	0.92	1.00	0.97	**0.96**

The CNN model outperforms all three SVM models, achieving the highest accuracy across all classes, and particularly improving the performance on the negative class. Among the SVM classifiers, the highest accuracy is achieved using the full set of 9 AUs detected by PyAFAR INFANT. The model that uses features extracted with PyA-

FAR ADULT shows the lowest performance, particularly struggling with the negative affect classification.

6 CONCLUSION

We evaluated the use of the PyAFAR INFANT Action Unit (AU) detector for classifying infant affect. An initial analysis of AU occurrence patterns across affect classes showed results consistent with previous work, confirming that PyAFAR INFANT is capable of capturing known AU patterns in infant facial expressions. We defined a baseline AU activation profiles: high activation of AU6 and AU12 in positive expressions, and AU3, AU4 and AU20 for negative expressions. However, manual inspection revealed certain limitations: confusion between AU12 (smiling) and AU20 (lip stretch), and under-detection of brow lowering (AU3 and AU4), particularly in subtle or ambiguous expressions.

A Support Vector Machine (SVM) model using these extracted AUs as features, achieved a high classification accuracy of 0.86, but the performance was weaker for the negative class (0.76). A detailed misclassification analysis revealed that incorrect predictions occur from subtle or ambiguous expressions, for co-activation of AUs from different affect classes, or for poor AU detection (especially brow-lowering). These issues limited in particular the model's ability to distinguish between positive and negative affect, when most of the AUs are activated.

We also compared performance of SVM using AUs from the PyAFAR INFANT and ADULT model. The infant model outperformed the adult model, especially on negative class, with 0.71 accuracy compared to 0.53 for adults. This reinforces the importance of using models trained on infants.

To address limitations of the AU-based classifier, we fine-tuned a Convolutional Neural Network (CNN) model pre-trained on AffectNet dataset for emotion recognition. With saliency maps, we validated that the model focused on meaningful facial regions, like eyes, mouth and brows, when making predictions. The CNN model significantly outperformed the SVM model, with an overall accuracy of 0.96, 0.10 points over the SVM.

The number of negative images misclassified as positive was significantly reduced with the CNN model. However, subtle negative expressions, which often lack strong AU activation, were still the main challenge and the primary reason for incorrect classification as neutral affect. This type of misclassifications occurred in three of the six misclassified samples by the CNN, which were the same cases already misclassified by the SVM. These errors are mostly explained by the inherent ambiguity of infant facial expression themselves.

To conclude, our findings demonstrate that the PyAFAR INFANT tool can be used efficiently for the recognition of infant affect. However, a deep learning approach using transfer learning achieved substantially higher performance. Deep learning approaches proved to be more effective than AU based models, particularly in handling ambiguous and subtle expressions.

6.1 LIMITATIONS

We are aware that our study is subject to several limitations. Firstly, they stem from the City Infant Faces dataset [Webb et al., 2017]. We only used greyscale images, which prevents analysis of the temporal dynamics of infant facial expressions. The dataset is small, it only consists of 154 images that mostly features white infants. The pictures are mainly frontal in nature, so there can be a lack of diversity in poses. These are some factors from our dataset that raise concerns about sampling bias and limit generalizability to broader populations with broader range of ethnicities, for example.

Similar concerns have been identified in both of the databases on which PyAFAR INFANT is trained on - MIAMI [Hammal et al., 2015a] and CLOCK [Bian et al., 2023]. The MIAMI dataset is described as ethnically diverse, but exact distribution is unclear, and CLOCK dataset also mostly contains white and hispanic infants in the study.

Additionally, we relied on automatic AU occurence predictions from PyAFAR INFANT. As we do not have ground-truth AU scores or manual FACS annotations for our dataset, it makes it more difficult to assess our results objectively.

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